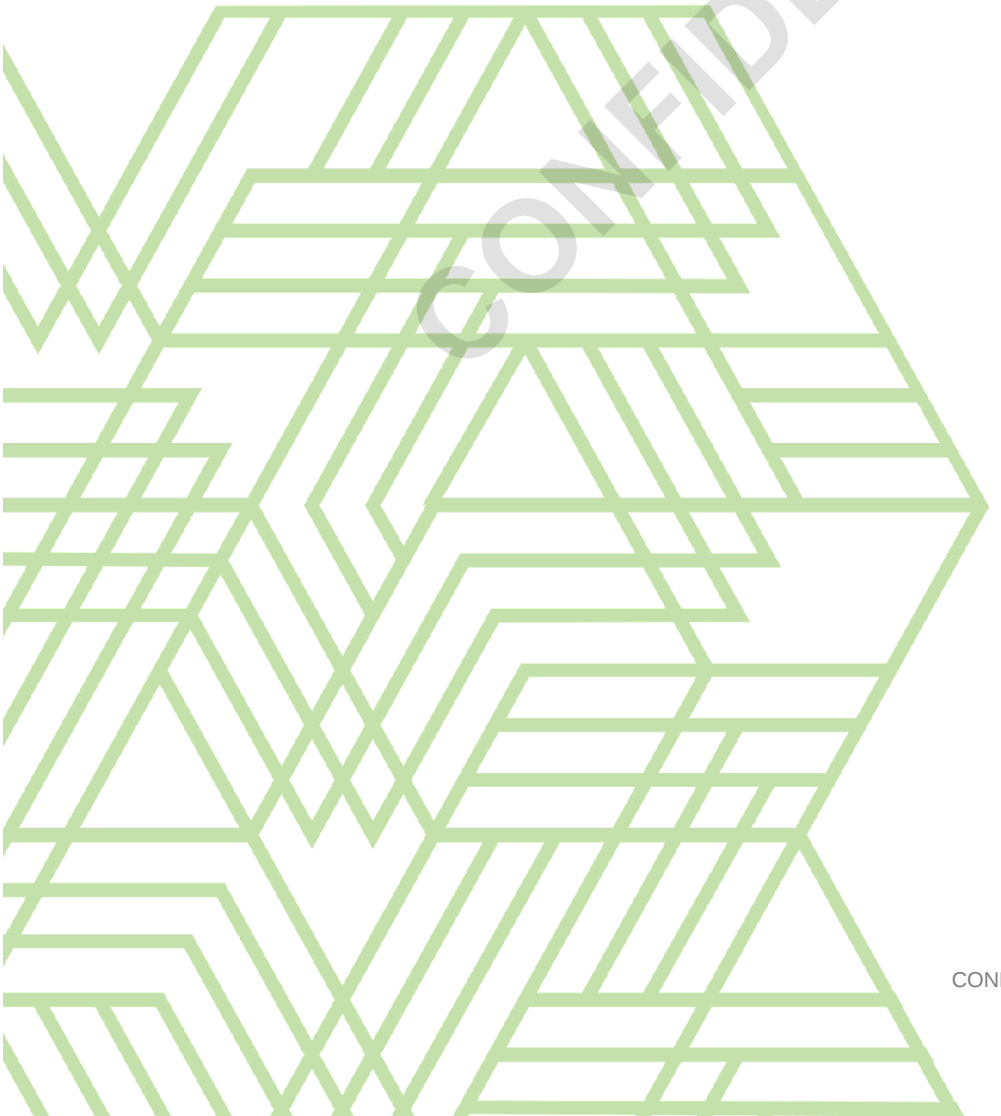
Using Secure Debug for ATM33/e

Application Note

SUMMARY: This document describes the Secure Debug operation of the ATM33/ATM33e Wireless SoC Series.

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Table of Contents

Acronyms and Abbreviations **2**
1. Introduction **3**
1.1 Supported Hardware and Software 3
2. Hardware Overview **3**
2.1 Debug Security Levels 4
2.1.1 Debug Enabled 4
2.1.2 Debug Secured 4
2.1.3 Debug Disabled 4
3. Software Overview **4**
3.1 Configuration 5
3.1.1 Compiling in the Public Key 5
3.2 Usage 5
3.3 Authentication Protocol 6
3.3.1 Authentication Commands 7
References **8**
Revision History **9**

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Acronyms and Abbreviations

Acronyms	Definition
ATM33/e	ATM3325 ATM3330 ATM3330e
ECC	Elliptic Curve Cryptography
ECDSA	Digital Signature Verification Algorithm
OTP	Over-The-Air
PEM	Privacy Enhanced Mail
SDK	Software Development Kit
SoC	System-on-Chip
SWD	Serial Wire Debug

1. Introduction

Secure Debug is a collection of hardware and software features to limit access to the debug port for devices in production. A simple hardware state machine controls the security level of the debug port (Serial Wire Debug or SWD) based on the value of a pair of OTP bits. The debug authentication library provides side channel authentication over a UART that can be used to unlock the debug port if it is configured in the secured state.

***Note**: Caution must be taken when configuring and testing secure debug. Once the OTP bits are configured to enable debug security they cannot be reverted. Most tooling will no longer function with the chip, and further development on the chip will not be feasible.*

1.1 Supported Hardware and Software

Secure Debug features are supported by the ATM33/e SoC series, and Atmosic SDK 5.4.0.1 or later. Please refer to the Atrmosic Customer Support Portal for latest Atmosic SDK release packages and release notes.

2. Hardware Overview

At a hardware level, the debug security state at power-on is defined by two OTP bits (ATM_OTP_MASK_SEC_DBG_DEBUG_SECURED and ATM_OTP_MASK_SEC_DBG_DEBUG_DISABLED). Additionally There is a register SECURE_DEBUG_STAT that facilitates the allowed state transitions at runtime.

Hardware applies the debug security state prior to the CPU booting. No intervention is required by software to enforce the security state. When secure debug is enabled, access through SWD is disallowed even if benign boot is enabled. When the port is secured (rather than disabled), the state can be cleared by software after a software challenge to prove the identity of the debug access requester.

The software components require a UART bus to communicate the authentication challenge and response. Therefore, UART_RX and UART_TX will need to be available. The implementation can be demonstrated on UART0 that is readily available for applications.

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2.1 Debug Security Levels

2.1.1 Debug Enabled

This is the default configuration. If neither OTP is set the debug interface will be enabled and freely available at all times. The debug port cannot be disabled by software in this mode.

2.1.2 Debug Secured

If ATM_OTP_MASK_SEC_DBG_DEBUG_SECURED is set the debug interface will be disabled by hardware at powerup. The SECURE_DEBUG_STAT register can enable the debug interface (sec_debug_unlock) or force the debug interface to stay disabled (sec_debug_sticky_disable) until the next power on sequence. This functionality is available in the sec_debug driver.

Once this mode is enabled, most *'make'* recipes provided by the Atmosic SDK will cease to function and use of the debug interface will be severely limited based on the method of authentication. This mode is only recommended for devices in production that are ready to be sealed/packaged for the final end user.

Secure DEBUG may become incompatible with certain production workflows. The restrictions imposed come at a cost and are intended to prevent tampering with devices in most settings. Please keep this in mind when this feature is used.

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2.1.3 Debug Disabled

If ATM_OTP_MASK_SEC_DBG_DEBUG_DISABLED is set the debug interface is permanently disabled and cannot be enabled.

3. Software Overview

Debug Authentication is implemented at the software level. The SDK provides an authentication side channel operating over UART that can perform authentication using elliptic curve cryptography (ECC) digital signature verification algorithm (ECDSA).

Debug authentication only has an effect if the device is in Debug Secured mode. Authentication is still possible in the other modes for testing and simulation purposes, but the hardware state of debug security will not be able to change.

3.1 Configuration

Secure debug runs as part of MCUboot. All examples can be built with secure debug enabled by adding the following build time options:

USE_MCUBOOT=1 USE_SECURE_DEBUG=1 <SECURE_DEBUG_KEY=”pem file” >

- USE_MCUBOOT=1 is used to enable MCUboot in our boot chain.
- USE_SECURE_DEBUG=1 is used to enable secure debug
- SECURE_DEBUG_KEY=<pem file> is an optional variable to choose a specific .pem file. A .pem file is a public/private key certificate file generated by tools such as openssl. The default uses the MCUboot development public key certificate which is widely distributed. Note that the Secure Debug key and the Secure Boot key should not be the same.

Do not use the default PEM file in production. This is only for development purposes and widely distributed in SDK releases.

3.1.1 Compiling in the Public Key

By default the public key is compiled into the MCUBoot image. As part of the build process, a public_debug_keys.gen.c file is generated and included in the MCUBoot image. From there, MCUBoot can use the key for debug authentication.

3.2 Usage

Debug authentication happens during the boot sequence while MCUBoot is executing. If no authentication occurs during the boot sequence, MCUboot will sticky lock the debug port until the device is reset.

Authentication is always possible, even if the OTP bits have not been set. This allows authentication to be tested while debug is still available.

The SDK provides a simple python script to authenticate with the device. It can be invoked in the following manor:

sec_debug_unlock.py -k <SECURE DEBUG PEM FILE> -p <SERIAL PORT> [-b <BAUD RATE>]

- -k <SECURE DEBUG PEM FILE> pem file containing the private key

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- -p <SERIAL PORT> serial port to authenticate on
- -b <BAUD RATE> Optional baud rate, defaults to 115200

After authentication is successful the device will disable the watchdog and enter a benign state, ready for the debugger to attach.

3.3 Authentication Protocol

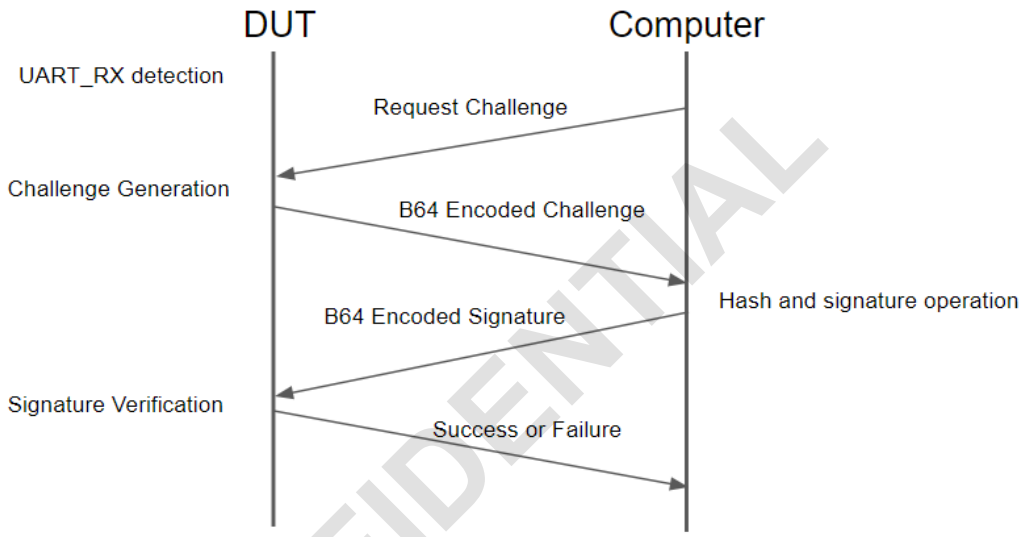


Figure 1 - Authentication Flow

The authentication protocol over UART is a simple challenge/response. The process starts when the UART_RX pin is held high during boot. When this is detected there is a default timeout of 5 seconds. If no valid command is received before the timeout expires, boot continues as normal and the debug port will be sticky disabled.

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3.3.1 Authentication Commands

All commands begin with “DBG “ and end with a newline character “\n”. All responses from the device are plain text and end with a newline character “\n”.

Command	Effect	Text Response from Device
DBG STATUS	None	If authentication has failed or has not been attempted ● Debug locked
DBG REQUEST	Generates a new challenge	Challenge: <BASE64 ENCODED CHALLENGE>
DBG RESPONSE <BASE64 ENCODED SIGNATURE>	Decodes the signature and attempts to validate the signature. If the signature is valid, debug is unlocked. If it is not valid, a new challenge will need to be generated.	● If the signature is not valid ○ “Signature decode bad” ● If the signature is not valid ○ “Bad signature” ● If the signature valid ○ “Debug Unlocked”

After a device has been unlocked it will no longer respond to any commands.

Example exchange:

Sending:	'DBG REQUEST\n'
Received:	'Challenge: Hhvcb0GenPBp6a2qfM8jKdfhiD+DEGbomKqAKSf+8Sb1qJ3qFhevPZ47KXBnDe9J45877Tkbm6IN1NiRg9cWNQ==\n'
Sending:	'DBG RESPONSE k+3BG4u3JmNOM25NKHW+pEG/dWGeVR368SuwJDfSJIiThh+5E1uroW6pbD9EZIHJaDcEN3xC1TwrdbPYl5InOg==\n' Generates a new challenge
Received:	'Debug unlocked\n'

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References

Title	Document Number	Link
Atmosic SDK Quick Reference Guide	ATM-QRGSDK	Atmosic Support website, submit a support request for access credentials if needed.
IDE Auxiliary Flash Programming Tool User Guide	ATM-UGIDEFPT	Atmosic Support website
MCUboot bootloader design	N/A	https://docs.mcuboot.com/design.html

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Revision History

Date	Version	Description
March 17, 2023	0.50	Initial version created for SDK 5.4.0.1 release.

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www.atmosic.com